

Project #14: Smart Steps: Designing and Developing a Microprocessor-Controlled Prosthetic Ankle for Enhanced Mobility in Amputees

Microprocessor-Based Adaptive Damping for a Transtibial Prosthetic Ankle-Foot System

PROJECT AIM: Develop a compact, low-cost prosthetic ankle that senses gait and terrain, identifies user intent, and adjusts damping in real time to improve stability, comfort, and mobility.

1. Project Overview and Motivation

This graduation project focuses on the design and development of a smart microprocessor-controlled prosthetic ankle-foot system for people with transtibial (below-knee) amputation. Aligned with Egypt's Vision 2030 for inclusive healthcare and local technology development, the project addresses the limitations of conventional passive feet, which generally provide fixed mechanical behavior and cannot adapt adequately to changes in walking speed, slopes, stairs, uneven ground, or transitions between sitting and standing.

The proposed device combines a compact ankle-foot structure with adaptive hydraulic or magnetorheological damping, real-time sensing, embedded control, safe battery management, and connected tuning software. Unlike a purely passive foot, the proposed semi-active system changes its resistance according to gait phase and operating conditions. The aim is to provide controlled plantarflexion and dorsiflexion, improved shock absorption, stable rollover, and safer movement without requiring the high energy consumption of a fully powered ankle.

2. Principal Objectives

- Design a compact and mechanically robust ankle-foot unit suitable for the available build height of below-knee amputees.
- Develop an adaptive damper and actuation mechanism that changes ankle resistance during stance and swing.
- Fuse IMU, ankle-angle, load, pressure, and temperature measurements for gait-phase and terrain recognition.
- Implement embedded control, BMS, safety monitoring, wireless communication, and clinician-oriented tuning software.

System Architecture of the Smart Microprocessor-Controlled Prosthetic Ankle

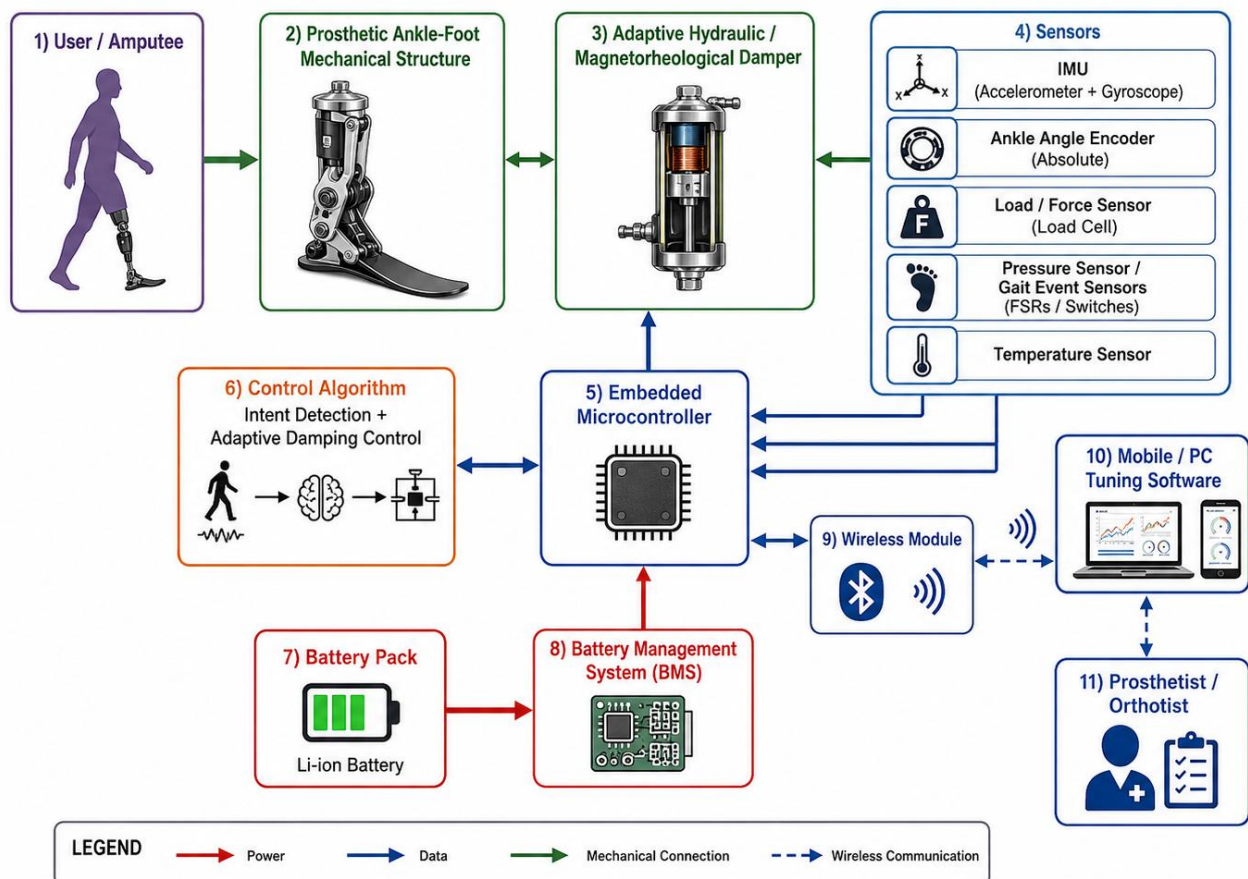


Figure 1. Proposed architecture integrating the compact ankle-foot mechanism, adaptive damping, sensors, embedded control, power management, and clinician interfaces.

3. Engineering Design and System Integration

Mechanical and adaptive damping design. The ankle-foot structure will be developed using lightweight, durable, and locally manufacturable materials such as aluminum alloys, stainless steel at highly loaded joints, engineering polymers, and a composite energy-storage foot plate. The vertical build height will be minimized so the unit can fit within the anatomical space available between the residual limb socket and the ground. The damping module may use a motor-controlled hydraulic valve or a magnetorheological damper to provide low resistance during swing and appropriately higher, phase-dependent resistance during loading response, mid-stance, terminal stance, and controlled descent.

Sensors and embedded electronics. A socket or shank IMU measures acceleration and angular velocity; an ankle encoder measures joint angle; load cells or strain-based force sensing estimate axial load and ankle moment; optional plantar-pressure sensors detect heel strike and toe-off; and temperature sensing supports thermal protection. An ARM-class microcontroller acquires and filters the signals, performs sensor fusion, detects gait events, estimates terrain and user intent, and calculates the required damping command within a real-time loop.

Power, communication, and safety. The electronics are supplied by a rechargeable lithium-ion battery with a BMS for cell balancing, over-current and over-temperature protection, state-of-charge estimation, and safe charging. Bluetooth Low Energy or an equivalent wireless link connects the ankle to a PC, tablet, or smartphone. Watchdog logic, mechanical motion limits, sensor plausibility checks, fail-safe damping, low-battery warnings, and event logging are included to support reliable operation.

Connected Software, Embedded System, and Power Management for the Smart Prosthetic Ankle



Figure 2. Connected embedded system, sensing, power management, charging, safety functions, clinician dashboard, and user mobile application.

4. Software and User Interfaces

The prosthetist dashboard will support patient records, alignment and calibration, gait-mode selection, live ankle-angle display, damping-level adjustment, battery status, control-parameter tuning, diagnostic alerts, and export of test data. The mobile application will provide simplified monitoring and quick mode selection for level walking, ramps, and stairs. Settings that affect safety or clinical alignment should remain protected by role-based access, while the user receives clear status, battery, and fault information.

5. Adaptive Damping Control Strategy

The control system is organized as a closed-loop sequence: sensing, feature extraction, gait-phase and intent detection, damping decision, actuator command, ankle response, and performance feedback. During initial contact and loading response, the controller provides controlled plantarflexion and impact attenuation. During mid-stance and terminal stance, damping is adjusted to support stable progression and energy-efficient rollover. During pre-swing and swing, resistance is reduced to permit toe clearance and smooth repositioning. Terrain, walking speed, measured load, and safety constraints are used to modify the damping setpoint.

The first prototype can begin with transparent rule-based or finite-state control because it is interpretable and easy to validate. Subsequent development may compare fuzzy logic, model-based adaptation, or machine-learning classifiers for gait and terrain recognition. The final controller must prioritize deterministic timing, bounded commands, safe-state behavior, and the ability to revert to a stable default damping level whenever sensor quality or communication is degraded.

Adaptive Damping Control Workflow for the Smart Prosthetic Ankle

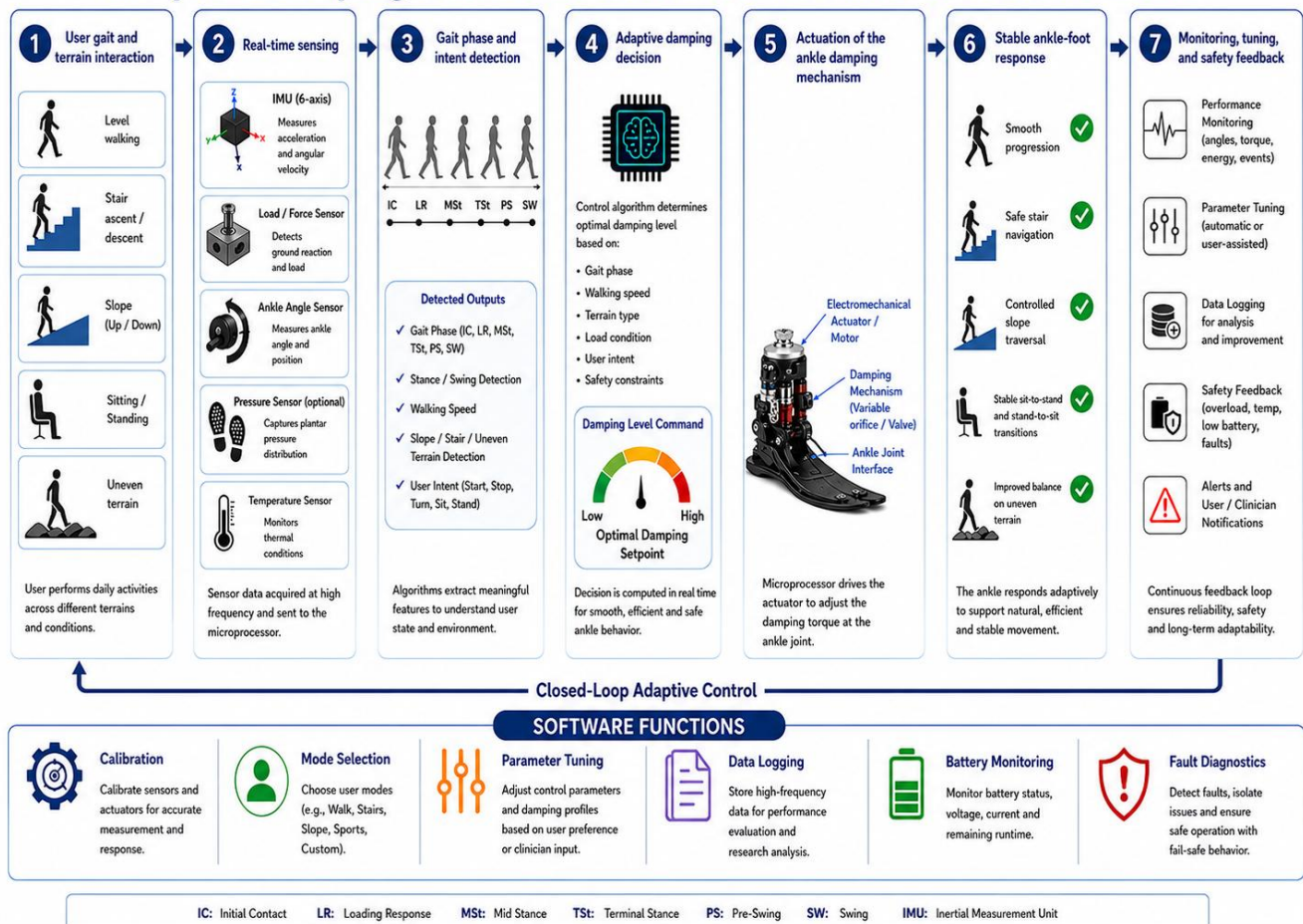


Figure 3. Closed-loop adaptive damping workflow from user interaction and real-time sensing to safe ankle response, monitoring, tuning, and diagnostics.

6. Implementation, Testing, and Expected Deliverables

Validation and Performance Evaluation

- CAD/FEA checks, mechanical bench tests, damping characterization, and endurance tests.
- Sensor calibration, controller timing, gait-event accuracy, and wireless reliability.
- Level walking, ramps, stairs, uneven terrain, and sit-to-stand scenarios using a safe test protocol.
- Metrics: ankle angle, damping response, load, stability, energy use, battery life, and fault recovery.

Expected Project Deliverables

- Compact ankle-foot mechanical prototype and adaptive damping module.
- Embedded controller, sensor interface, battery pack, BMS, charger, and safety circuits.
- Real-time firmware, adaptive damping algorithms, data logging, PC dashboard, and mobile app.
- Engineering drawings, source code, test results, user manual, and final technical report.

7. National Relevance and Project Impact

A locally designed prosthetic ankle can reduce dependence on expensive imported devices, strengthen collaboration between engineering and rehabilitation medicine, and create a platform for future clinical and industrial development in Egypt. The project will train students in mechanical design, embedded systems, sensing, control, power electronics, software engineering, biomechanics, safety, and responsible testing while targeting a practical assistive technology that can improve mobility, independence, and quality of life.

Online Video Resources

Video 1: [Watch Video 1](#)
Video 2: [Watch Video 2](#)

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